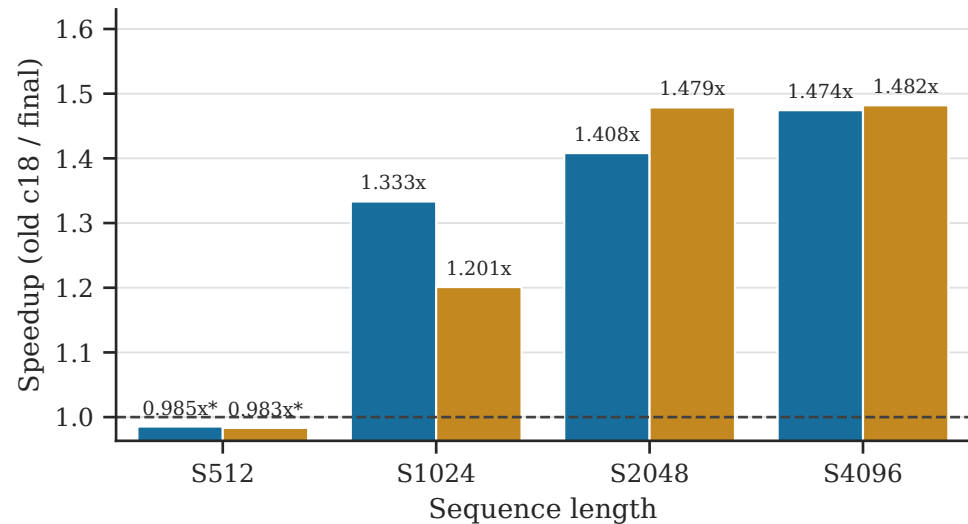


FP16 Forward Speedup — Final vs Old c18

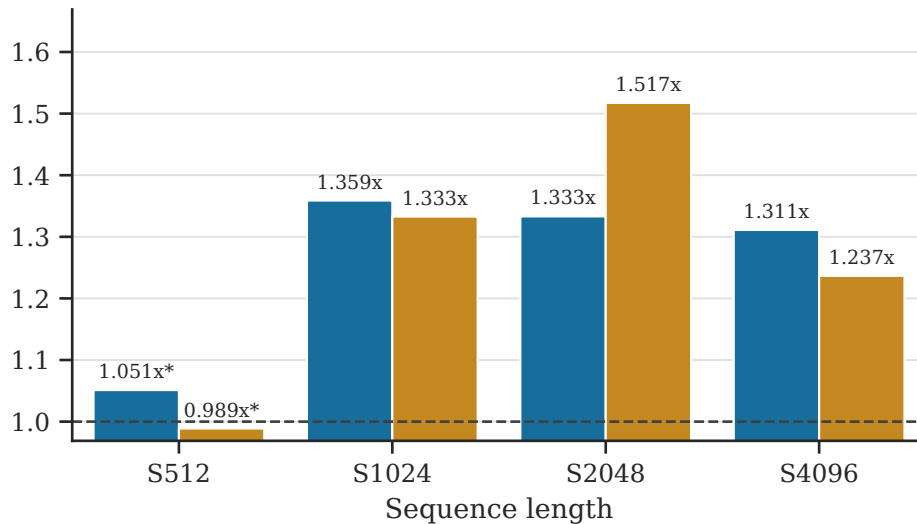
* Gated short-sequence fallback; panels use independent y-scales

Non-causal Causal

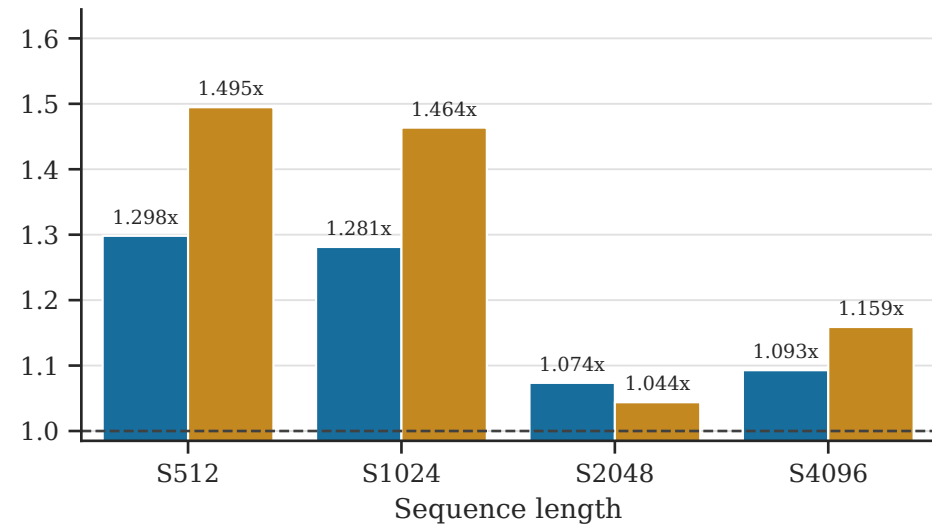
Head dimension D=64



Head dimension D=128



Head dimension D=256



Median across per-position 10%-trimmed means; 1000 warmups, 100 trials, 3 ABBA rounds (NC D128/S4096: 5-round recheck)